# SIEM Log Monitoring and Security Event Analysis using Elastic SIEM

## 1. Introduction

In today's digital environment, organizations generate a large volume of logs from systems, applications, and network devices. Monitoring these logs is essential to detect suspicious activities and security incidents. A Security Information and Event Management (SIEM) system helps Security Operations Center (SOC) teams collect, analyze, and correlate logs to identify potential threats.

The objective of this project is to design and implement a **beginner-level SOC lab** using **Elastic SIEM** to collect Windows event logs, monitor authentication activities, manually detect suspicious login behavior, and investigate security events following a SOC analyst workflow.

This project focuses on **log monitoring, investigation, and reporting**, which are core responsibilities of a Tier-1 SOC analyst.

## 2. Objectives of the Project

The main objectives of this project are:

- To understand the working of a SIEM platform
- To collect Windows event logs using Winlogbeat
- To establish a baseline of normal authentication behavior
- To simulate suspicious login activity
- To manually detect potential brute-force login attempts
- To investigate security events using Kibana
- To document findings in a structured and professional manner

# 3. Lab Environment and Tools Used

## 3.1 Hardware and Platform

- Host Operating System: macOS
- Virtualization: Virtual Machine environment

## 3.2 Software and Tools

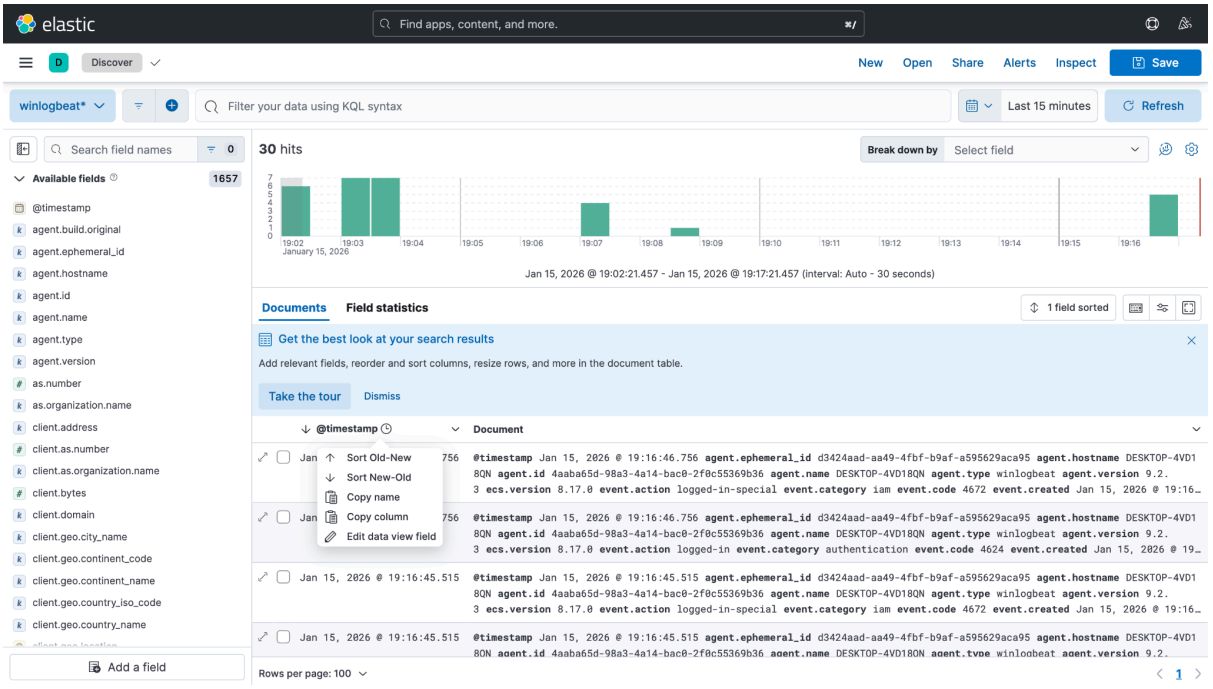
- **Elastic Stack**
    - Elasticsearch (log storage and indexing)
    - Kibana (log visualization and analysis)
- **Winlogbeat**
    - Used for forwarding Windows event logs
- **Windows Virtual Machine**
    - Acts as the log source

# 4. Lab Architecture

The lab environment is designed with a simple architecture suitable for a beginner SOC setup.

## Architecture Description:

- A Windows virtual machine generates security events.
- Winlogbeat is installed on the Windows system to collect event logs.
- Winlogbeat forwards logs to Elasticsearch.
- Kibana is used to visualize, search, and analyze the collected logs.




# 5. Log Collection Setup and Verification

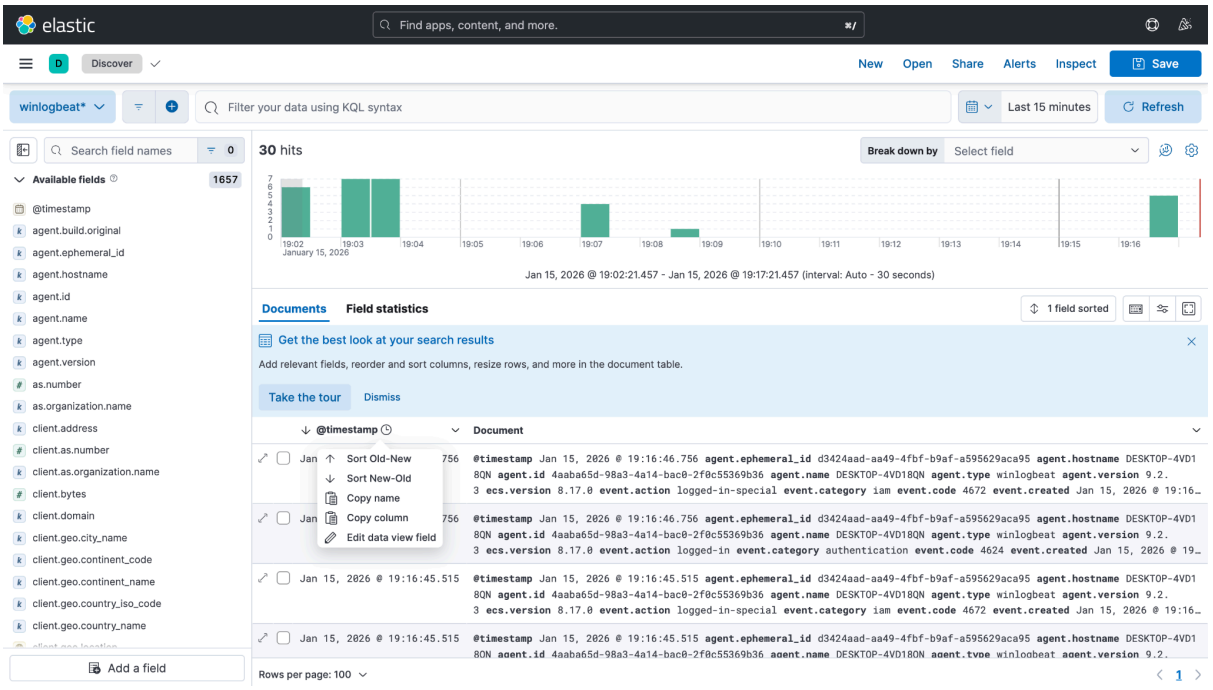
After installing and configuring Winlogbeat on the Windows virtual machine, log forwarding to Elastic SIEM was verified.

## Verification Steps:

- Opened Kibana and navigated to the Discover page
- Selected the data view `winlogbeat-*`
- Verified continuous incoming events

## Observations:

- Windows Application, System, and Security logs were visible
- Event timestamps updated in real time
- Log ingestion was functioning correctly




# 6. Baseline Authentication Activity (Normal Behavior)

Before analyzing suspicious behavior, it is important to understand what **normal login activity** looks like.

## Action Performed:

- A successful login was performed on the Windows system

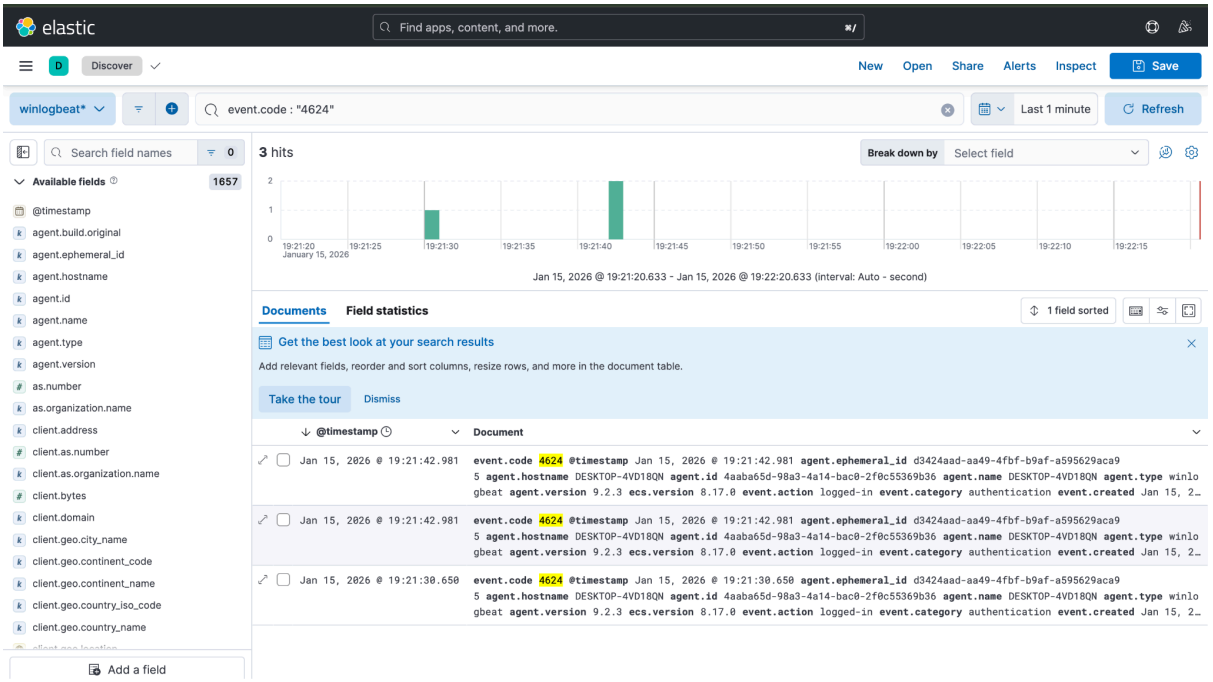
## Detection Query Used:

```
event.code:4624
```

(Event ID 4624 represents successful authentication events.)

## Observations:

- Successful login events were recorded
- User account, hostname, and timestamp fields were present
- This established a baseline for normal behavior




# 7. Attack Simulation – Failed Login Attempts

To simulate suspicious activity, multiple failed login attempts were intentionally generated.

## Action Performed:

- Incorrect passwords were entered multiple times on the Windows login screen

## Purpose of Simulation:

This activity simulates:

- Password guessing attacks
- Brute-force login attempts

No real system damage was caused, as this was a controlled lab environment.

# 8. Manual Detection of Suspicious Login Activity

After generating failed login attempts, the SIEM was used to manually detect suspicious activity.

## Detection Query Used:

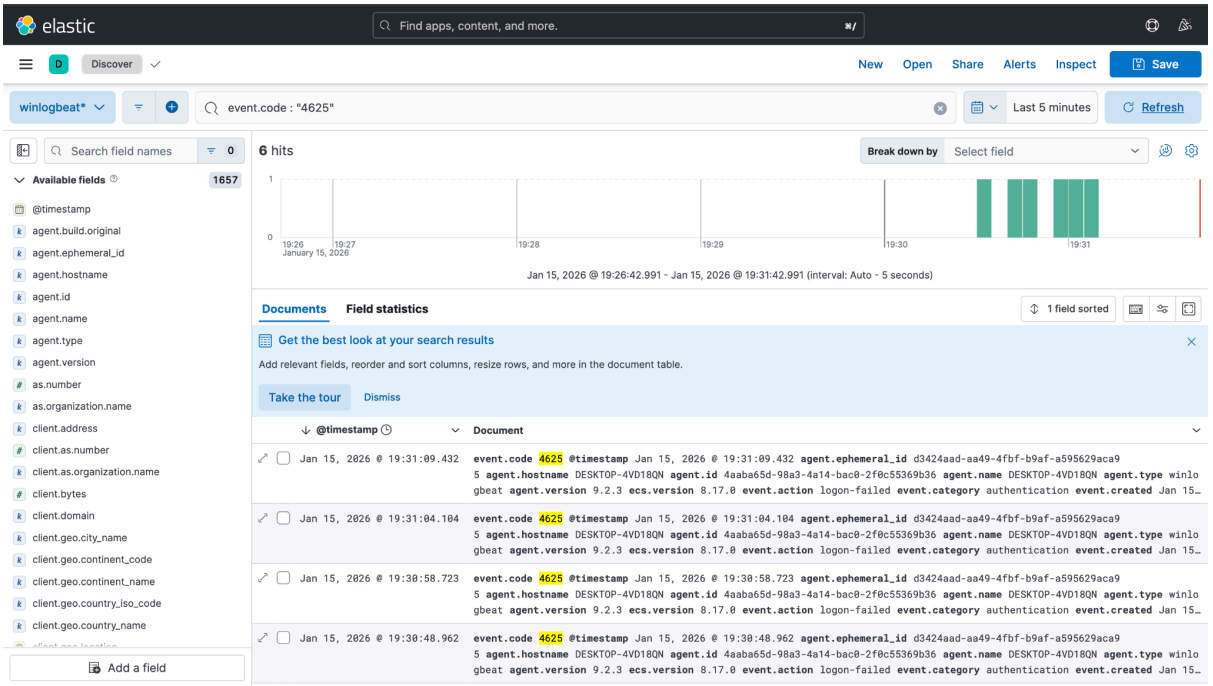
```
event.code:4625
```

(Event ID 4625 represents failed authentication attempts.)

## Observations:

- Multiple failed login events were observed
- The same user account was targeted repeatedly
- Events occurred within a short time window
- No successful login followed the failed attempts

This pattern indicates potential brute-force behavior.



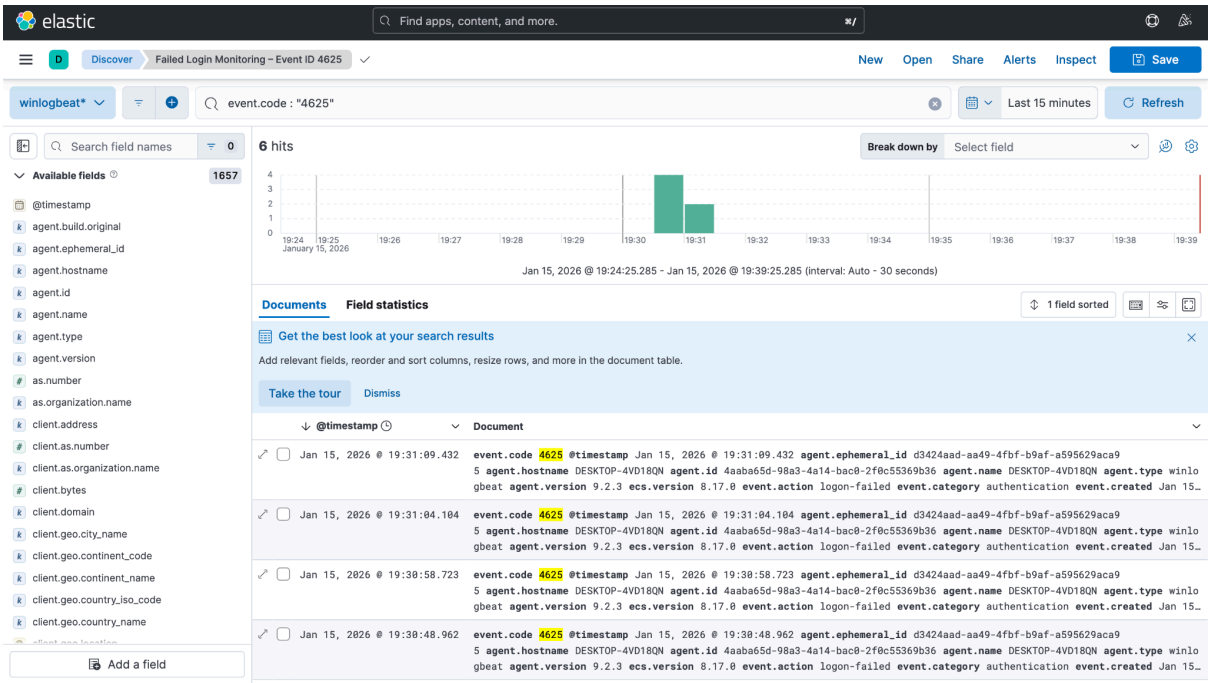
# 9. Saved Monitoring View

To support continuous monitoring, a saved search was created in Kibana.

### Saved Search Details:

- **Name:** Failed Login Monitoring – Event ID 4625
- **Purpose:** Quick access to failed authentication events

Saved searches allow SOC analysts to monitor recurring security patterns without repeating manual queries.




# 10. Investigation of Failed Login Events

Each failed login event was analyzed in detail to understand the nature of the activity.

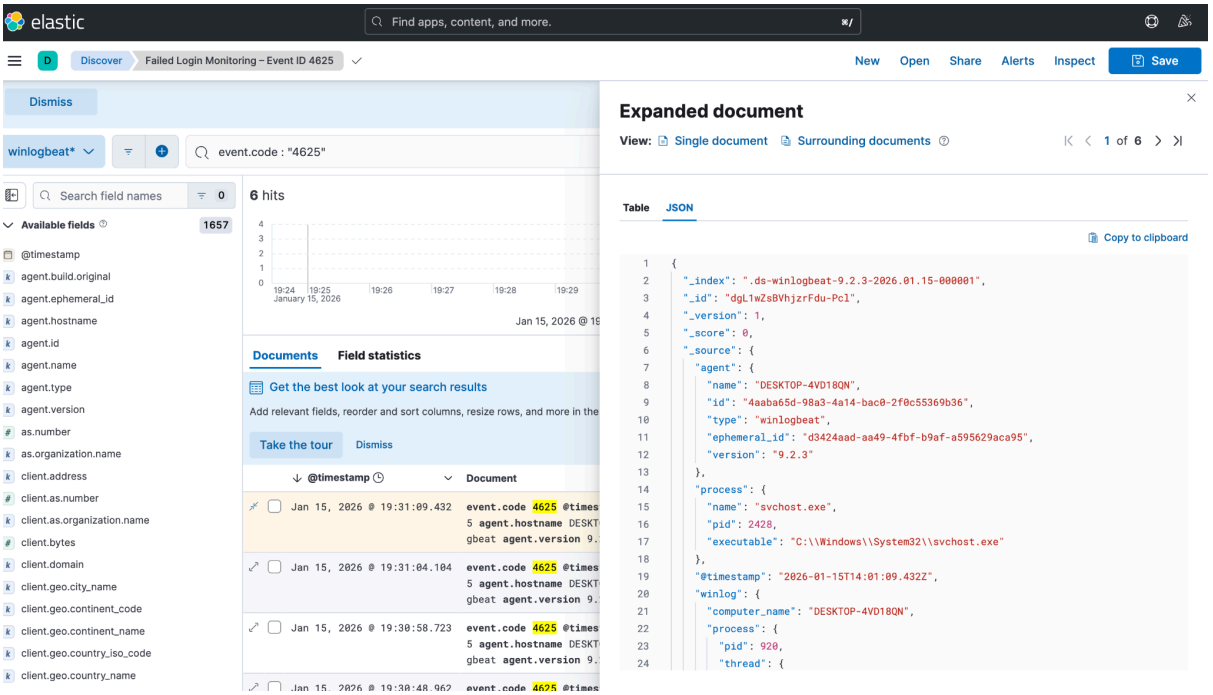
### Fields Examined:

- Username
- Hostname
- Timestamp

- Frequency of login failures

### Investigation Findings:

- Repeated authentication failures targeting a single user
- Activity occurred over a short duration
- No evidence of successful account compromise




# 11. Analysis and Classification

Based on the investigation, the observed activity was analyzed and classified.

### Classification:

- **Attack Type:** Brute-force login attempt (simulated)
- **Source:** Internal (lab environment)
- **Severity:** Medium

### Analysis Statement:

The pattern of multiple failed authentication attempts within a short time window for the same user account is consistent with brute-force login behavior. No successful authentication events were observed after the failed attempts, indicating that the attack was unsuccessful.

# 12. Limitations of the Project

- Automated SIEM alerting was not enabled due to lab constraints
- Detection was performed manually using log analysis
- The environment represents a simplified SOC setup

Despite these limitations, the project effectively demonstrates core SOC analyst skills.

# 13. Learning Outcomes

Through this project, the following skills were gained:

- Understanding SIEM concepts and workflows
- Log collection and verification
- Manual threat detection
- Event investigation and analysis
- SOC-style documentation and reporting

# 14. Conclusion

This project successfully demonstrated the use of Elastic SIEM to collect and analyze Windows authentication logs. By simulating suspicious login activity and manually detecting failed authentication events, a SOC analyst workflow was followed from detection to investigation and analysis. The project provided practical exposure to real-world SOC tasks such as log monitoring, incident analysis, and documentation.